



3 Torsion

Course: Strength of Materials

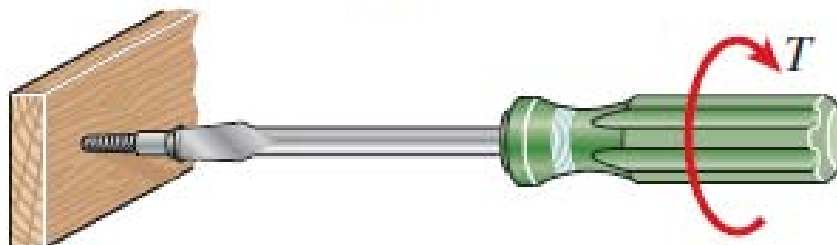
Academic Year: 2017-2018

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Introduction



(a)



(b)

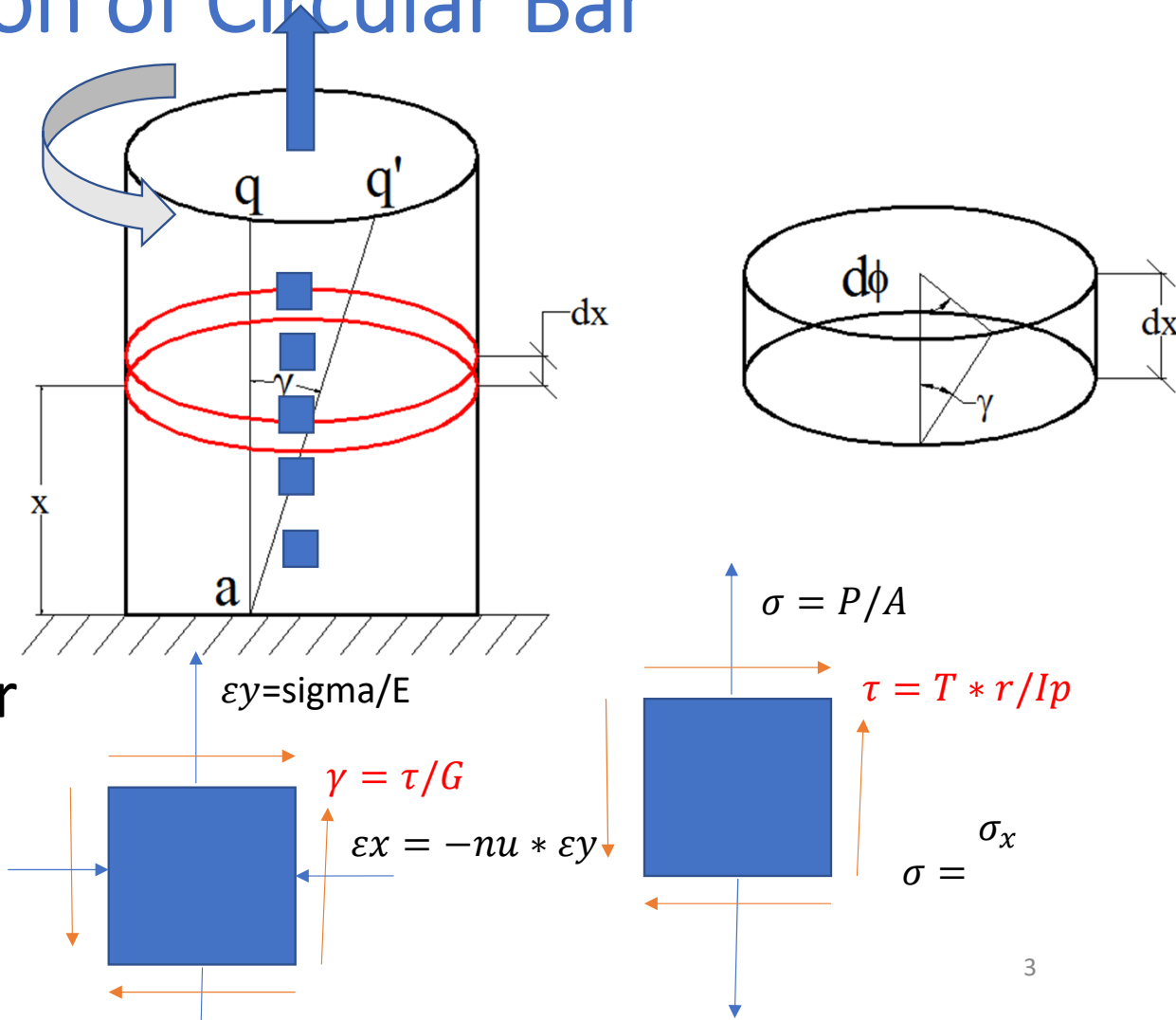
Torsional Deformation of Circular Bar

Shear Strain at the Outer Surface

- $\gamma_{max} = \frac{r d\phi}{dx}$
- $\theta = \frac{d\phi}{dx}$ (Twist angle)
- $\gamma_{max} = r\theta$

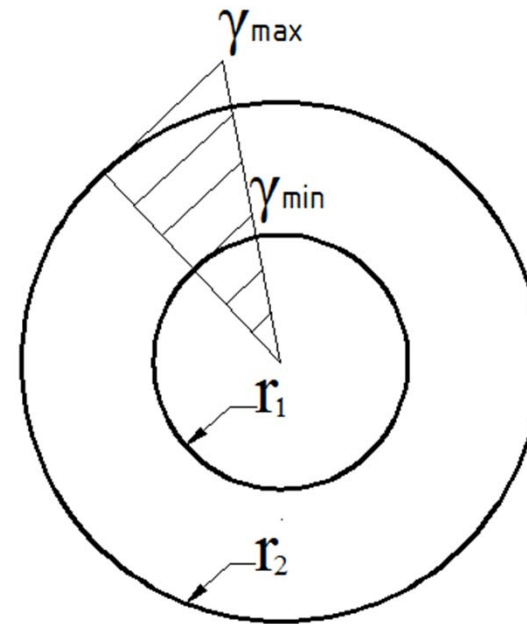
Shear Strain Within the Bar

- $\gamma = \rho\theta = \frac{\rho}{r} \cdot \gamma_{max}$



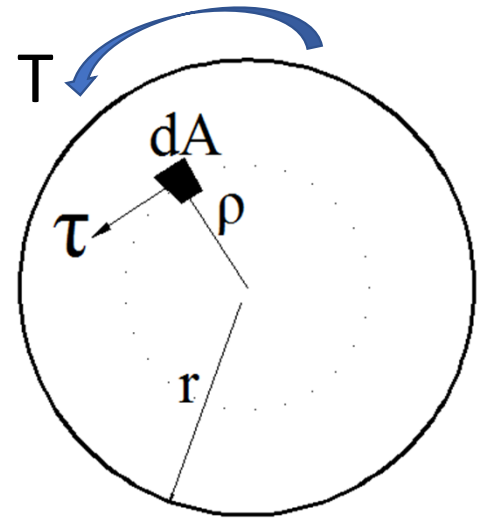
Torsional Deformation of Circular Tube

- $\gamma_{max} = r_2 \frac{\phi}{L}$;
- $\gamma_{min} = \frac{r_1}{r_2 \gamma_{max}} = r_1 \frac{\phi}{L}$



Circular Bar of Linearly Elastic Materials

- $\tau = G\gamma$
 - $\tau_{max} = Gr\theta; \tau = G\rho\theta = \frac{\rho}{r} \cdot \tau_{max}$
 - $dT = \rho\tau dA = \frac{\tau_{max}}{r} \rho^2 dA$
- $$\Rightarrow T = \int dT = \int \frac{\tau_{max}}{r} \rho^2 dA$$
- $$\Rightarrow T = \frac{\tau_{max}}{r} \int \rho^2 dA = \frac{\tau_{max}}{r} I_p$$
- $$\Rightarrow \tau_{max} = \frac{T}{I_p} r$$



Circular Bar of Linearly Elastic Materials

- For a circular bar

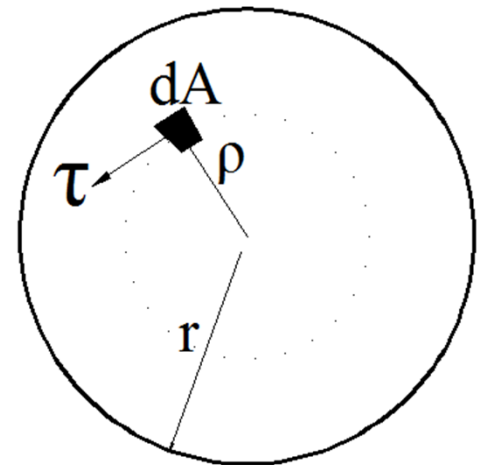
$$I_p = \frac{\pi d^4}{32} = \frac{\pi r^4}{2}$$

$$\Rightarrow \tau_{max} = \frac{32T}{\pi d^4} \cdot \frac{d}{2} = \frac{16T}{\pi d^3}$$

- For a circular tube

$$I_p = \frac{\pi}{2} (r_2^4 - r_1^4) = \frac{\pi}{32} (d_2^4 - d_1^4)$$

$$\tau_{max} = \frac{32T}{\pi (d_2^4 - d_1^4)} \cdot \frac{d_2}{2}$$



Angle of Twist

- $\theta = \frac{T}{GI_p} \left(= \frac{d\phi}{dx} \right)$
- For Pure Torsion $\phi = \theta L$
 $\Rightarrow \phi = \frac{T.L}{GI_p}$
- Torsional stiffness and flexibility are given, respectively:
 $k_T = \frac{GI_p}{L}; f_T = \frac{L}{GI_p}$



Angle of Twist

For Uniform Torsion $\phi = \theta L$

$$d\phi = \frac{T}{GI_p} dx \Rightarrow \int_{\phi_1}^{\phi_2} d\phi = \int_{x_1}^{x_2} \frac{T}{GI_p} dx$$

- Torsional stiffness and flexibility are given, respectively:

$$k_T = \frac{GI_p}{L}; f_T = \frac{L}{GI_p}$$



Angle of Twist

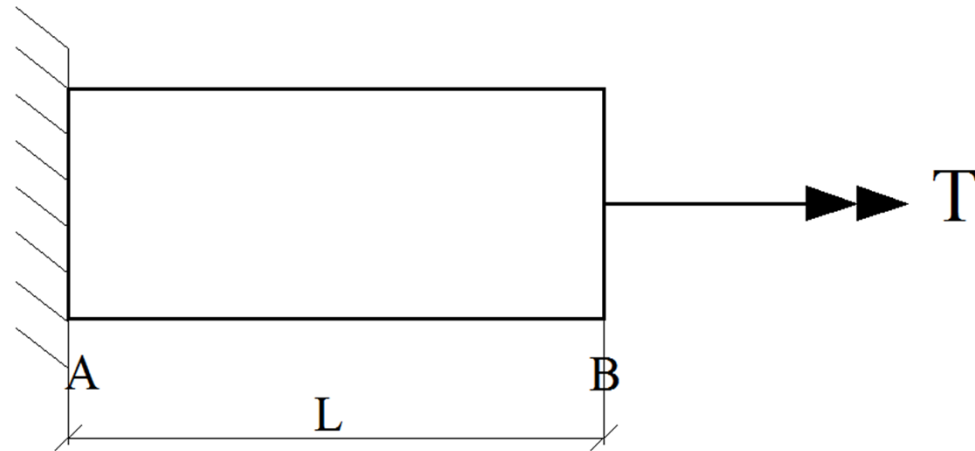
For General Torsion $\phi = \theta L$

$$d\phi = \frac{T}{GI_p} dx \Rightarrow \int_{\phi_1}^{\phi_2} d\phi = \int_{x_1}^{x_2} \frac{T}{GI_p} dx$$

- Case 1: $T = \text{cte}$, $I_p = \text{cte}$

$$\phi_B - \phi_A = \frac{T}{GI_p} (L - 0)$$

$$\phi_B = \frac{T}{GI_p} L$$



Angle of Twist

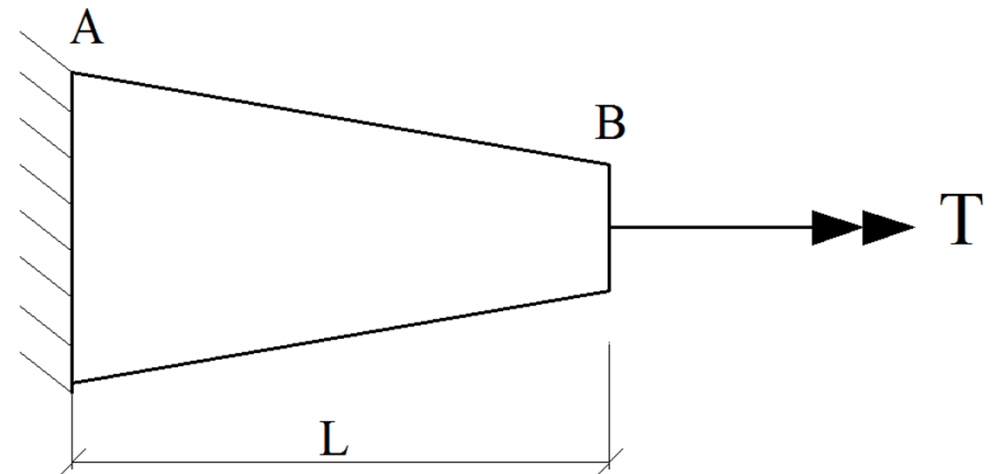
For Normal Torsion $\phi = \theta L$

$$d\phi = \frac{T}{GI_p} dx \Rightarrow \int_{\phi_1}^{\phi_2} d\phi = \int_{x_1}^{x_2} \frac{T}{GI_p} dx$$

- Case 2: $T = \text{cte}$, $I_p \neq \text{cte}$

$$\phi_B - \phi_A = \frac{T}{G} \int_0^L \frac{dx}{I_p(x)}$$

$$\phi_B = \frac{T}{G} \int_0^L \frac{dx}{I_p(x)}$$



Angle of Twist

For Normal Torsion $\phi = \theta L$

$$d\phi = \frac{T}{GI_p} dx \Rightarrow \int_{\phi_1}^{\phi_2} d\phi = \int_{x_1}^{x_2} \frac{T}{GI_p} dx$$

- Case 3: $T \neq \text{cte}$, $I_p \neq \text{cte}$

$$\phi_B - \phi_A = \frac{1}{G} \int_0^L \frac{T(x) dx}{I_p(x)}$$

$$\phi_B = \frac{1}{G} \int_0^L \frac{T(x) dx}{I_p(x)}$$

